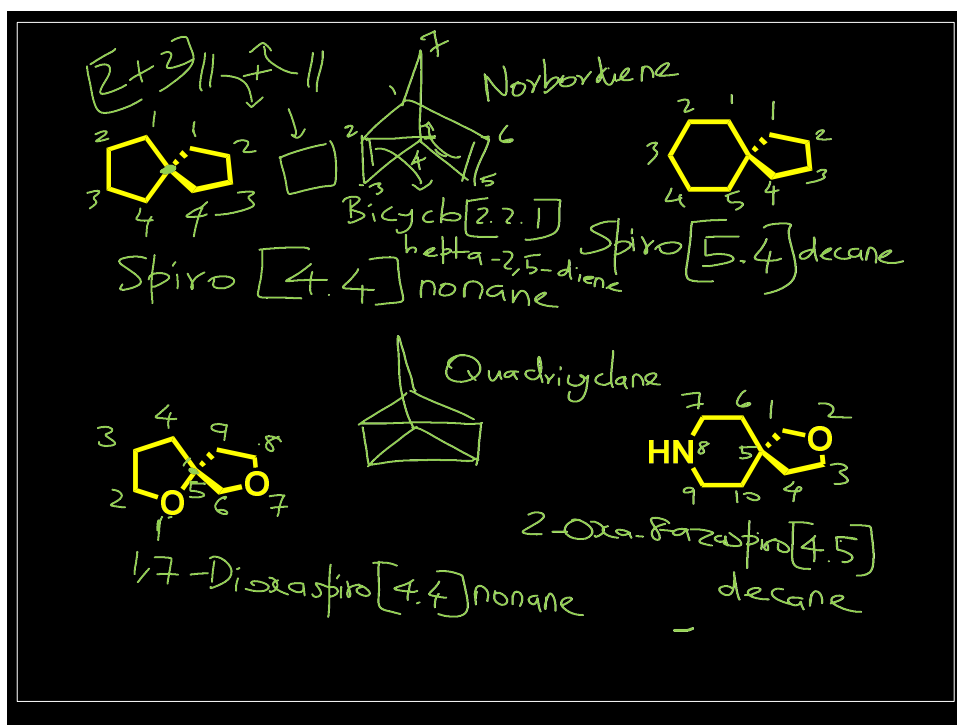
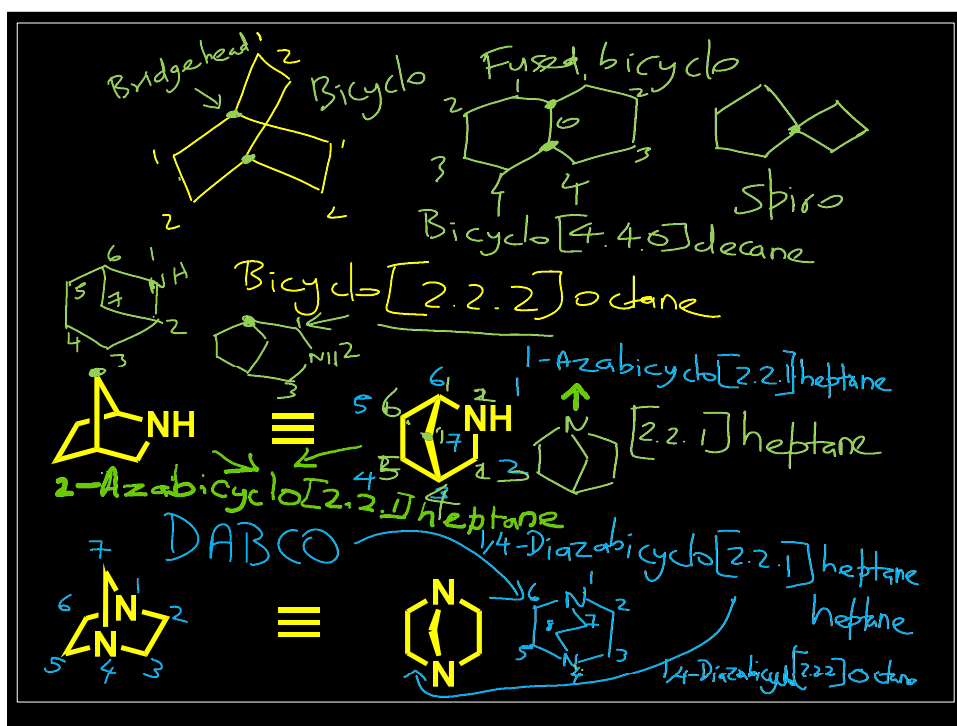




(4*S*,5*E*)-4,6-dichlorohept-5-en-2-one

hept(a)	parent (heptane)
en(e)	unsaturation ending
di	multiplicative prefix
2 4 5 6	locants
one	suffix for principal characteristic group
chloro	substituent prefix
E S	stereodescriptors
()	enclosing marks

Fig. 1: Criteria for choosing the senior parent compound.

Pure Appl. Chem. **2020**; 92(3): 527–539

Concepts related to organic chemistry

① Inductive effect

②

Organic compounds

Structure

↙ ↘

Stability ↔ Bonding

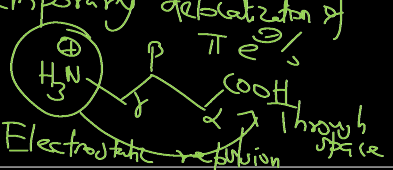
Ph $\text{=N}^{\oplus}\text{=N}^{\ominus}$ $\text{O}-\text{O}^{\ominus}$

Reactive intermediates

Synthesis

↕

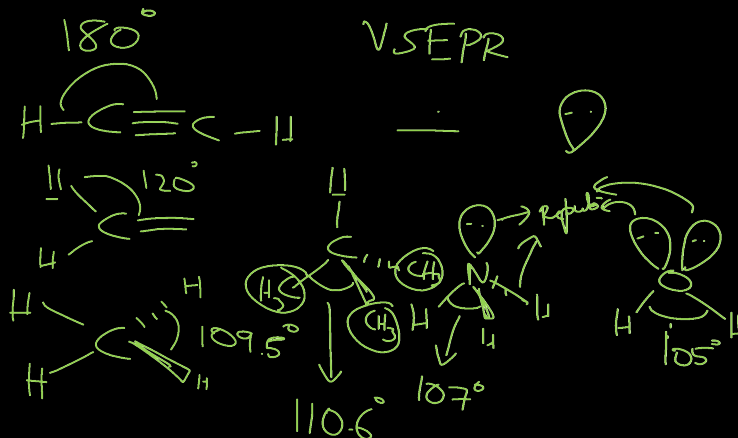
Reactivity

- ① Inductive effect $C \rightarrow Cl$
Delocalization of σe^- s
 - ② Mesomeric effect / Resonance
Delocalization of πe^- s
 - ③ Hyperconjugation
Delocalization of σ & πe^- s
 - ④ Electromeric effect
Temporary delocalization of πe^- s
 - ⑤ Field effect
 - ⑥ Steric effect $\rightarrow$ Electrostatic repulsion
- 

Bonding

Covalent bonding

VSEPR



Hybridization

Orbital mixing

⇓

Directionality

